

A possible outcome

Over time there is a finite probability that a radioactive particle will decay, but the decay event itself is impossible to predict.

Tunneling may provide a solution as to why particles sometimes, but not always, escape from an atomic nucleus.

TUNNELING THROUGH

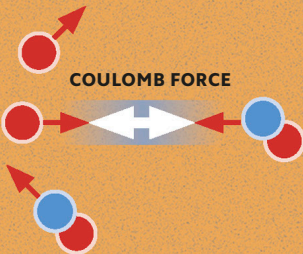
The amplitude of a wave function decreases exponentially across a potential barrier, but it can still end up with a nonzero value on the other side of the barrier, and therefore there is a nonzero chance of the particle being found on the other side of the barrier.

BARRIER

NUCLEI IN THE SUN

The sun's core has lightweight hydrogen nuclei, which are subject to electromagnetic coulomb forces: similarly charged particles repel each other and differently charged particles attract.

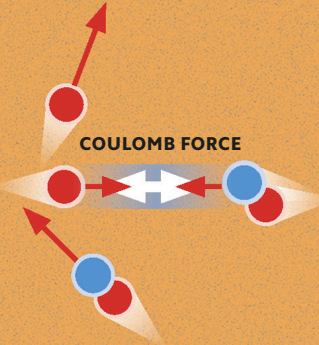
COULOMB FORCE



OPPOSING FORCES

Temperatures of millions of degrees give nuclei huge kinetic energy, but conditions inside the sun are not hot enough to account for the vast number of particles that overcome repulsion.

COULOMB FORCE



NUCLEAR FUSION

There are many more fusion reactions in the sun than would be expected given the coulomb barrier in place, and so quantum tunneling must play a key role.

